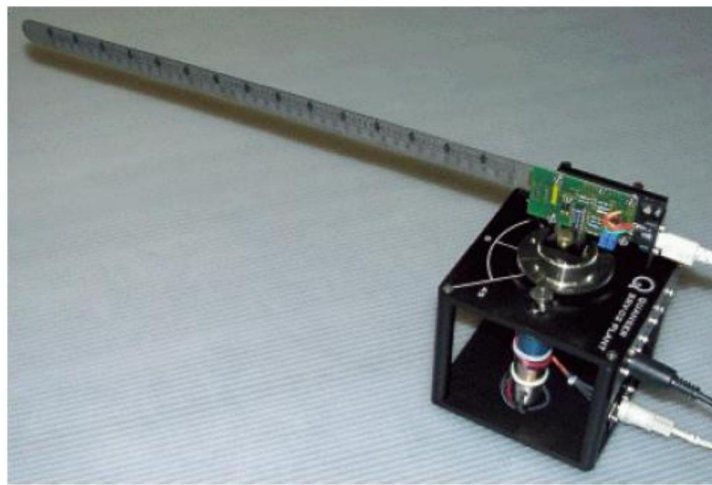


Control of Cyber-physical Systems

LABORATORY WORK REPORT — PART I

Identification of a Flexible Robot Arm Joint



Shift L04

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2025/2026 – 1st Quarter

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Introduction

1 Question 1 - Analyzing the problem (2.5 pts)

1.1 Why is achieving this control objective nontrivial? (2.5 pts)

Our control objective is to manipulate the motor so that the tip of the flexible bar precisely follows a specified angle. A trivial approach would be to create a lookup table of electric voltages for the DC motor, that would ensure a fixed position of the bar tip. However, there are a few standout reasons why this control objective is nontrivial and why that approach would not be feasible. The voltage applied to the DC motor determines a torque, which in turn produces a speed. So, a constant voltage does not correspond to constant position, as it requires an integrator to get from speed to position. Furthermore, this speed is not reliably constant.

Let's picture the scenario where we apply a constant voltage to the motor:

When the voltage, and in turn, the torque is applied, it takes a short time interval for the speed to increase, due to the gearbox friction and elastic torque from the bar (non minimum phase zero). Then, the speed converges, as the bar keeps spinning around in one direction, but its value is not reliably constant, due to external disturbances. So the position value is approximately linear but drifts with time.

What might the approximate open-loop pole distribution of the system look like?:

- **One real pole on the origin:** Integrator. Conversion of motor speed to position
- **One real positive zero:** Non minimum phase zero. Bending of the bar in the opposite direction, before moving as desired
- **Two complex pole pairs, on the left-half plane:** To represent the two bending modes that affect the underdamped motion of the tip of the flexible bar. The pole pair for the fundamental bending mode (whole bar flexing like an arch) has lower frequency and higher damping, while the second bending mode (section of the bar bending the opposite way) has higher frequency and lower damping.

Would a simple feedback controller, consisting of a single gain amplifying the tracking error, effectively address the problem? Closing the loop with a single gain, would move the pole on the origin towards the non minimum phase zero, destabilizing the system. It might also excite the complex pole pairs, leading to unstable oscillations.

2 Question 2 - Interfacing the plant with the computer (5 pts)

2.1 Comment on the motor operation. Explain in particular why their angular position never stabilizes when its command signal is constant in the open loop conditions described. (0.5 pts)

As mentioned previously, there's an integrator between the angular position and the voltage applied to the DC motor. This means that a constant command signal results in a continuous rotation.

2.2 Explain the procedure carried out to perform sensor calibration. (4.5 pts)

2.2.1 Potentiometer sensor calibration

Our objective is to find the constant K_p in the following equation 1 that describes the relation between θ_e (electrical tension provided by the sensor) and θ (actual deflection angle)

$$\theta = K_p \theta_e \quad (1)$$

To estimate the value of the constant K_p , we applied a step input of 1V to the DC motor 1, with the bar detached to minimize nonlinearities in the movement. From the potentiometer voltage measurement we get a sawtooth looking plot, given that the potentiometer's measured voltage repeats itself every whole rotation of the cursor. There is also a visible saturation around both limits of the voltage scale $[-5V, 5V]$, caused by the potentiometer dead band, so those samples are not representative and we won't include them in any computation.

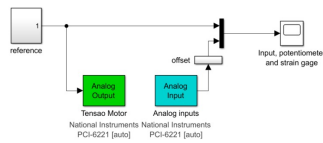


Figure 1: Simulink block diagram

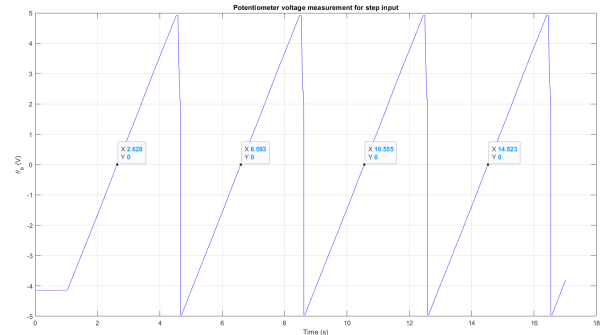


Figure 2: Potentiometer Voltage for step input

Provided we want to get the linear relation between θ_e and θ , we had to convert the time scale to angular position. To get a precise conversion, we took the zero intercept points for 3 periods of rotation. Knowing each rotation is 360° degrees, we plot these samples and fit a linear regression, obtaining the relation between time and angular position (angular speed) 3. The R2 score proves the reliability of this method as there is no significant variance between periods.

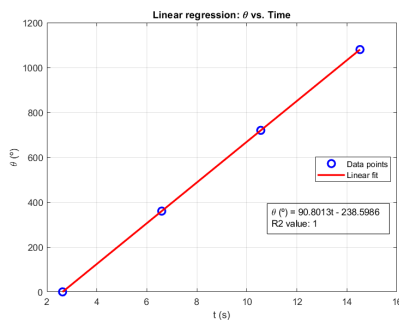


Figure 3: Linear regression: θ vs. Time

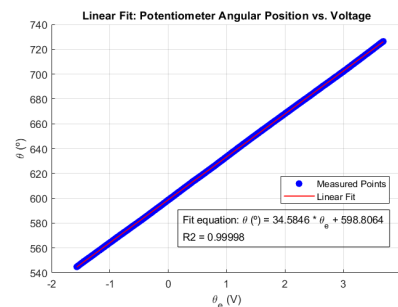


Figure 4: Linear regression: θ vs. θ_e

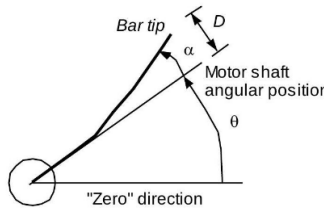
Finally, we picked an approximately linear interval of samples [6, 8](s), avoiding the potentiometer dead band, and converted the time values to angular position using our inferred angular speed ($\omega = 90.8013$ (°/s)). From here, we could fit another linear regression and get a final $K_p = 34.5846$ (°/V) 4

2.2.2 Strain gauge sensor calibration

This time, our objective is to find the constant K_e in the following equation 2 that describes the relation between α_e (electrical tension provided by the sensor) and α (actual deflection angle)

$$\alpha = K_e \alpha_e \quad (2)$$

To estimate the value of the constant K_e , we conducted 12 voltage measures, using the calibration "comb", each for a different deflection angle. The deflection angles were measured with the following approximation 3



$$\alpha = \frac{180}{\pi} \frac{D}{L} (\theta) \quad (3)$$

Figure 5: Schematic model of the flexible bar deflection.

As for the voltages, for each measurement, we let the value stabilize for about 5 seconds and took the average. Then we plotted the voltages against angles and did a linear regression 6, obtaining $K_e = 1.9213$ °/V

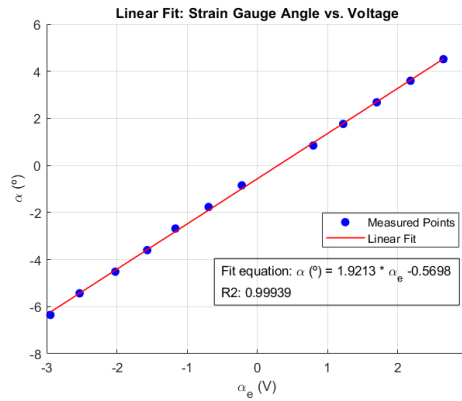


Figure 6: Linear Fit: Strain Gauge Angle vs. Voltage

3 Question 3 - Model identification and validation (9.5 pts)

3.1 Explanation on the tests performed on the plant to obtain the data used for identification. Discuss the results obtained with different types of excitation signals and the effect of very small amplitude and very large amplitude excitation signals. (2 pts)

During the laboratory sessions, several experiments were performed to obtain data for model identification. The plant was excited with different signals generated in Simulink, and the responses of the potentiometer and strain gauge were recorded for analysis. Each experiment was repeated to ensure reliability, with runs of about 20 seconds—long enough to capture several input cycles but short enough to avoid drift caused by motor asymmetries.

The tests mainly varied the frequency content of the excitation while keeping the amplitude fixed. PRBS inputs (0.3–0.5 Hz up to 0.3–3 Hz) excited a wide range of dynamics in a single run, providing rich data for identification. Sinusoidal inputs in the 0.3–0.5 Hz range showed phase lag and resonance effects, while square waves produced sharp transitions that emphasized transient dynamics and gearbox backlash. These three signals complemented each other: PRBS for broadband excitation, sinusoids for harmonic analysis, and square waves for highlighting nonlinear and transient effects.

Although frequency was the main focus, amplitude was also important. A value of 1 V was used consistently: smaller amplitudes could not overcome gearbox backlash, resulting in noisy signals with negligible motion, while larger amplitudes risked nonlinearities such as amplifier saturation, increased friction, or excessive bar deflection, potentially damaging the equipment.

3.2 Discussion of the sampling frequency. (1 pt)

The sampling frequency was fixed at 50 Hz, corresponding to a sampling period of 20 ms. This value was selected as a compromise between capturing the relevant dynamics of the plant and avoiding excessive sensitivity to noise. A higher sampling frequency would in principle allow the model to reproduce the system's dynamics with greater detail, but at the same time it would also increase the amount of high-frequency noise recorded by the sensors. Since the flexible joint does not present significant dynamics at very high frequencies, oversampling would mainly capture noise and generate redundant data, without a meaningful improvement in model accuracy.

Choosing a lower sampling frequency would risk aliasing and the loss of important dynamic information, especially in the frequency range where the bar and the motor interact. In that case, the identified model would fail to represent the system correctly. The frequency of 50 Hz was sufficiently high to avoid aliasing and to allow accurate identification, while not unnecessarily amplifying noise or producing excessive computational load.

3.3 Effect on identification of filtering the data. (1 pt)

The motor introduces an integrative effect, which makes the output signal drift when the input has a non-zero average. To remove this effect, the output was differentiated. Since differentiation amplifies high-frequency sensor noise, a first-order low-pass filter was applied to the differentiated signal, as suggested in the lab guide, with transfer function

$$H(z) = \frac{(1 - a_f)(1 - z^{-1})}{1 - a_f z^{-1}}.$$

The parameter a_f controls the strength of the filtering. Lower values of a_f remove more noise but risk attenuating relevant dynamics, while higher values preserve more of the signal dynamics but reduce less noise. The effect of different values of a_f is illustrated in the figures: with $a_f = 0.7$ the filter has little effect and most of the noise remains, while with $a_f = 0.9$ the signal becomes overly smooth and some dynamics are lost. The intermediate case of $a_f = 0.85$ provided the best compromise, strongly attenuating noise while preserving the main dynamics of the flexible bar. This filtered output was therefore used for the ARMAX identification, ensuring clean and representative data for the model.

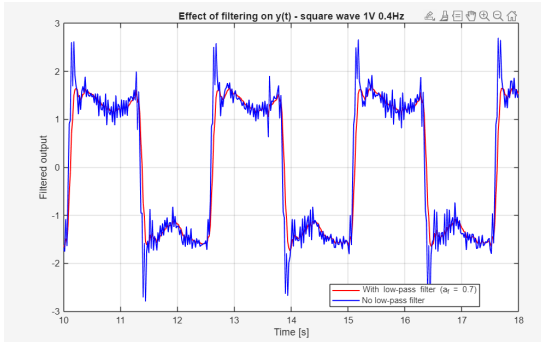


Figure 7: $a_f \approx 0.7$

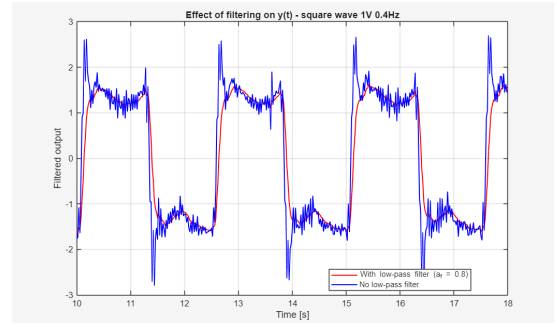


Figure 8: $a_f \approx 0.8$

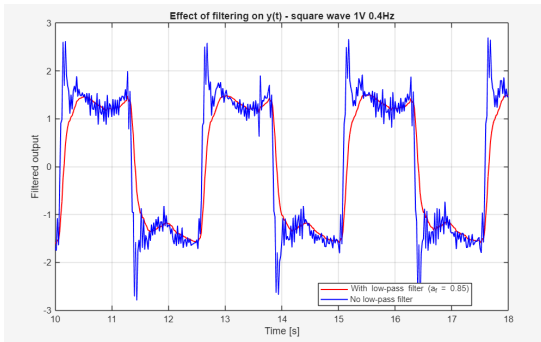


Figure 9: $a_f \approx 0.85$

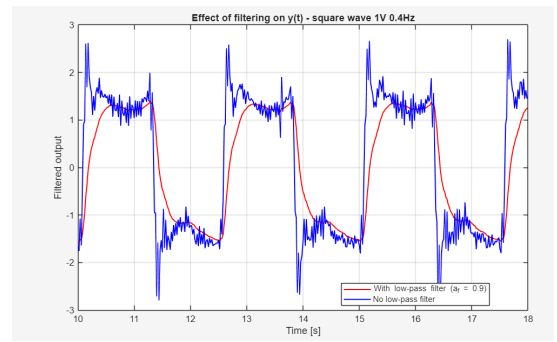


Figure 10: $a_f \approx 0.9$

3.4 Explanation on how the pole at the origin has been dwelt with. (0.5 pts)

The motor behaves like an integrator because its position is obtained by integrating the angular velocity. This creates a pole at the origin, which makes the output drift over time and complicates the identification.

To solve this, the output signal was differentiated so the integrator effect was removed, giving data that could be modeled more easily. Since differentiation increases high-frequency noise, a low-pass filter was added to smooth the signal while keeping the main dynamics. The ARMAX model was then identified using this filtered data.

Finally, the pole at the origin was added back into the model so that it correctly represents the real physical behavior of the motor.

3.5 Discussion on how the model orders have been decided. Take into consideration that the plant results from the interconnection of a DC motor and the flexible bar and discuss the models needed for each of them. (1 pt)

The model order reflects the plant's physical structure, comprising a DC motor, gearbox, and flexible arm. The motor behaves approximately as a first-order system, with its pole influenced by the electrical-mechanical time constant. Gearbox backlash, exacerbated by long-term use, and the gear ratio add effective inertia, modifying the motor dynamics. The flexible arm is governed by a partial differential equation whose solutions correspond to its vibrational modes. Considering the experimental sampling time and input frequencies, only the first two modes are observable. Initial motor movement causes the arm to bend oppositely, indicating a non-minimum phase zero. Consequently, the input voltage to tip speed model is expected to be third-order or higher with at least one non-minimum phase zero.

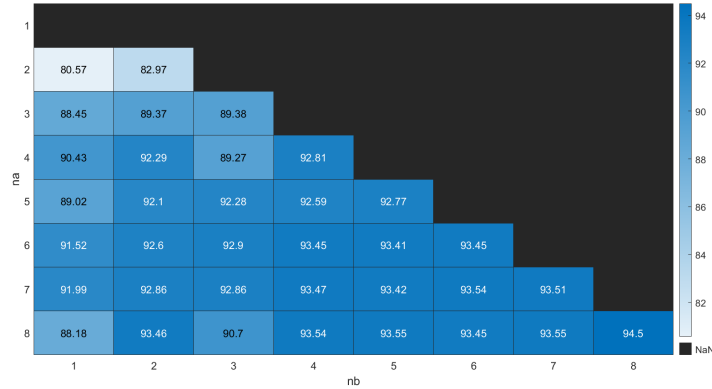


Figure 11: NRMSE for different ARMAX parameters

Following this, a MATLAB script was developed iterating over the possible $n_a = n_c$ and n_b values, computing an ARMAX model from that and comparing the obtained model with the training data based on the goodness of the fit between the simulated response and the output, measured data using the Normalized Root Mean Square (NMSE). For pseudorandom binary sequence with a bandwidth of 1 Hz. The results can be seen on the heatmap in figure 11.

Given the heatmap, it was decided to go for the combination $n_a = 5$ and $n_b = 2$. In fact, accounting for the first order model of the motor, and attaining to the fact that natural frequency poles translate into pairs of complex conjugate poles, that seems to be the best choice. Also, $n_b = 2$ suggests the existence of one zero, which we predicted would occur. Higher orders in n_a are not necessarily better, since they can lead to modeling unrepresentative physical characteristics of the plant. Also, a higher n_b would, in general, lead to an overfitting of the model to the training data, compromising its robustness.

3.6 Description of the final ARMAX model. (0.5 pt)

ARMAX stands for AutoRegressive Moving Average with eXogenous inputs. It estimates parameters allowing for the description of the system according to a difference equation structured as:

$$y(t) + a_1y(t-1) + \dots + a_{n_a}y(t-n_a) = b_1u(t-n_k) + \dots + b_{n_b}u(t-n_k-n_b+1) + c_1e(t-1) + \dots + c_{n_c}e(t-n_c) + e(t),$$

where n_a is the number of poles, n_b is the number of zeros plus 1, n_c is related to the white noise to be considered, and n_k is the number of input samples that occur before the input affects the output, also known as the pure delay.

As established in subsection 3.5, a good representation of the system is obtained using parameters $n_a = n_c = 5$, $n_b = 2$ and $n_k = 1$. This, applied to the PRBS signal and followed by the integrator that has to be added to the system, yields the coefficients that describe the system through a difference equation.

3.7 Description of the final state-space model. (0.5 pts)

The difference equation can then be translated to a state-space model:

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}u(k)$$

$$y(k) = \mathbf{C}\mathbf{x}(k) + \mathbf{D}u(k)$$

$$\mathbf{A} = \begin{bmatrix} 2.024 & -0.26 & -1.605 & 0.5323 & 0.6219 & -0.3126 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{C}^T = \begin{bmatrix} 0.07541 \\ -0.1461 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \mathbf{D} = 0.$$

3.8 Characterization of the plant open loop pole-zero plot, frequency response and time response of the model. (1.5 pt)

Having the state space representation of the system, it was possible to plot the Bode diagram and pole-zero map of the system.

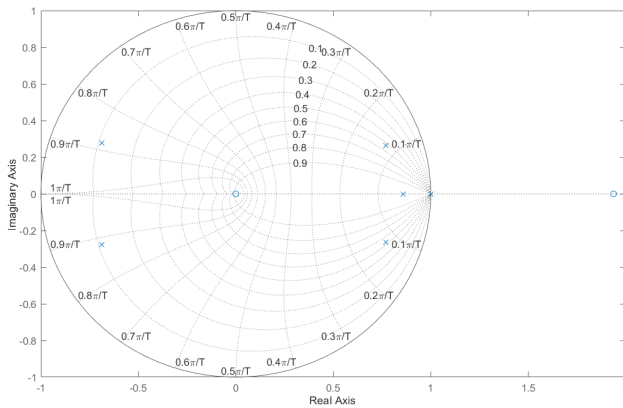


Figure 12: Pole-zero map

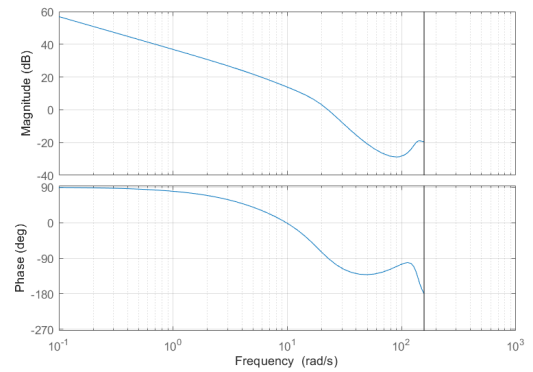


Figure 13: Bode Diagram

The pole-zero map can be readily interpreted and related to what was said in subsection 3.5. There are 2 pairs of complex conjugate poles. The poles with larger negative

real parts correspond to higher natural frequencies but exhibit lower damping coefficients compared to the other pair. Both, they represent the first two natural modes of the bar. The real pole near 1 can be related to the mechanical time constant of the motor and gearbox. The pole at 1 was manually introduced because of the integrator. Lastly, there are 2 zeros. The zero with positive real part is related to the non-minimum phase behaviour the system has. On the other side, the zero at the origin was also placed when the integrator was inserted in the model.

Looking at the bode plot, there isn't a clear sign of the first natural mode of the system, but around 10^2 rad/s there is a resonance corresponding to the second one. At low frequencies, the tip of the bar continues to rotate without settling, explaining the high gain in this region. At higher frequencies, the signal starts to be filtered by the plant, therefore the reduced gain.

3.9 Model validation. (1.5 pts)

Although not explicitly mentioned in the ARMAX model description, during lab sessions, it was a common practice to record multiple experiences on the plant with the same input signal. Then, each of this data was converted to an iddata object and merged to a higher-level iddata object that was the input of the ARMAX function. This way, the estimated model could be robust to random occurrences in each experiment, maximizing the likelihood of achieving a good representation of the system.

However, there was also some data that was recorded and not used to train any model. Therefore, the next plots show the validation of the obtained model by comparison to other system input-output responses. The following figure 14 depicts that for a PRBS $B = 3\text{Hz}$, a square wave and a sine wave, highlighting its robustness.

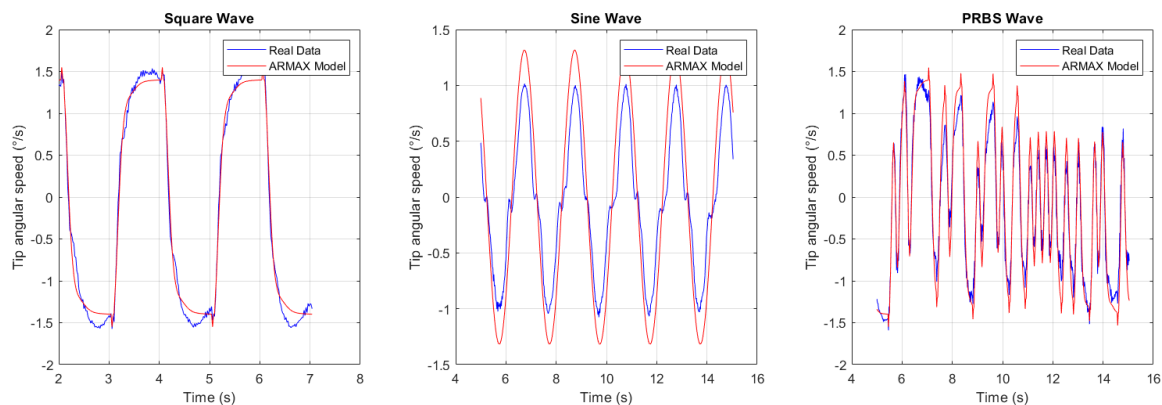


Figure 14: Validation Data

Conclusions

This laboratory focused on the identification of a model for the flexible robot arm joint. After calibrating the potentiometer and strain gauge sensors, experiments with different excitation signals were carried out to collect representative data. The signals were processed with differentiation and low-pass filtering to reduce noise and compensate for the motor's integrative effect, which enabled a more reliable identification.

From this data, ARMAX and state-space models were obtained, capturing the main dynamics of the system and showing good agreement with experimental validation. These results provide a robust representation of the plant and establish a solid foundation for the following control design stage.